

Prepare_GEPA.R

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Prepare_GEPA	<i>Pull gridded methane emissions maps for sectors not built with other functions from the Maasakkers gridded EPA inventory</i>
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Description

‘Prepare_GEPA’ writes 3 netcdf files of gridded methane emissions - 1 for thermogenic sectors, 1 for industrial landfills, and 1 for non-thermogenic sources.

Usage

```
Prepare_GEPA(  
  inventory_year,  
  input_directory,  
  output_directory,  
  plot_directory,  
  County_Tigerlines,  
  State_Tigerlines,  
  domain,  
  domain_template,  
  verbose  
)
```

Arguments

`inventory_year` Numeric indicating the desired year of data to use.

`input_directory` Character providing the full filepath to save/load input data

`output_directory` Character providing the full filepath to save processed data

`plot_directory` Character providing the full filepath to save figures. Only relevant if `verbose = TRUE`.

County_Tigerlines	SpatVector. United States Census Bureau county shapefile downloaded in Main.
State_Tigerlines	SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html . Only relevant if verbose=TRUE.
domain	SpatVector polygon outlining the desired output area
domain_template	SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system
verbose	Logical indicating whether to save visuals. It includes 3 plots of the gridded methane emissions, 1 for each netcdf file.

Details

This function downloads the most appropriate year of gridded methane data at 0.1 deg by 0.1 deg from Maasakkers et al. and pulls sectors that have not been calculated in other scripts. They are projected to the domain grid and combined into thermogenic and non-thermogenic groups, with the exception of industrial landfills which are saved separately. For thermogenic this includes

- mobile combustion
- Coal (abandoned, surface, and underground)
- Petroleum Systems (exploration, production, refining, and transport)
- Abandoned Oil and Gas
- Natural Gas (exploration, processing, and production)
- Industry (petrochemical, and ferroalloy)

And for non-thermogenic this includes

- Composting
- Enteric Fermentation
- Manure Management
- Rice Cultivation
- Field Burning

The data is available at <https://doi.org/10.5281/zenodo.8367082>. The closest year will be automatically downloaded.

See reference [Maasakkers et al.](#)

Value

Nothing is returned from the function, but the main outputs are 3 netcdf files of the methane emissions from the gridded EPA product. They are titled "GEPA_thermo.nc" for gridded EPA thermogenic, "GEPA_non_thermo.nc" for non-thermogenic, "GEPA_ind_landfill.nc" for industrial landfills.

Author(s)

Joe Pitt, <madeup@wisc.edu>

Kris Hajny, <blank@fake.edu>

Israel Lopez-Coto, <test@test.edu>

Examples

```
library(terra)
grid_bbox=cbind(c(-76.65,-73.65),c(38.97,40.97))
grid_res=0.01
grid_crs="epsg:4326"
grid <- rast(nrows=diff(range(grid_bbox[,2]))/grid_res,
             ncols=diff(range(grid_bbox[,1]))/grid_res, xmin=min(grid_bbox[,1]),
             xmax=max(grid_bbox[,1]), ymin=min(grid_bbox[,2]), ymax=max(grid_bbox[,2]),
             crs=grid_crs)
grid_vect <- as.polygons(ext(grid),crs=grid_crs)
Prepare_GEPA(inventory_year=2018,
             input_directory=~/.//Desktop/in/",
             output_directory=~/.//Desktop/out/",
             domain=grid_vect,
             domain_template=grid,
             verbose=TRUE,
             State_Tigerlines=vect("~/.//Desktop/in/State_Tigerlines/tl_2018_us_state.shp"),
             County_Tigerlines=vect("~/.//Desktop/in/County_Tigerlines/tl_2018_us_county.shp"),
             plot_directory=~/.//Desktop/plots/")
```

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